Algorithmen für KI und Data Science II
SoSe 2025

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Exercise Sheet 07

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Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. **Markov chain**

(a) Consider a discrete-time Markov chain with state space $\mathcal{S} = \{0, 1, 2, 3\}$ and the one-step transition probability matrix **2P**

$$P = \begin{pmatrix} 2q & p & 0 & 0 \\ 3p & q & 0 & 0 \\ 0 & \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \\ \frac{3}{5} & 0 & 0 & \frac{2}{5} \end{pmatrix}$$

(i) Determine p and q.

(ii) Draw the state diagram of the Markov chain.

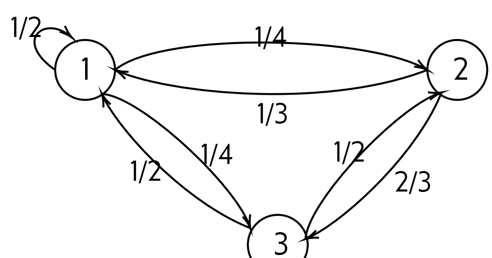
(b) Consider the Markov chain shown in Figure below. **3P**

(i) Is the chain irreducible?

(ii) Is the chain aperiodic?

(iii) Find the stationary state of this chain.

(iv) Is the stationary state unique?



(c) In Würzburg, München, and Nürnberg the University enrolls a lot of students. During this period, the distribution of students among these three cities followed certain patterns. In Würzburg, 80 percent of the people of Würzburg attended the local institution, while the remaining 20 percent went to Nürnberg. Among the people of Nürnberg, 40 percent attended the Nürnberg school, while an equal split of the remaining 60 percent went to Würzburg and München. As for the people of München, 70 percent attended the München institution, 20 percent went to Würzburg, and the remaining 10 percent chose Nürnberg. **3P**

(i) Find the probability that the grandson/grandnieces of a person from Würzburg went to Würzburg.

(ii) Modify the above by assuming that the people of a Würzburg always went to Würzburg. Again, find the probability that the grandsons/grandnieces of a person from Würzburg went to Würzburg.

2. **Eigenvalues and eigenvectors**

   (a) For the given matrix A find the eigenvector that belongs to the eigenvalue $\lambda = 2$ $\boxed{\text{1P}}$

$$A = \begin{pmatrix} 4 & 5 & -3 \\ -7 & -8 & 3 \\ 1 & -5 & 8 \end{pmatrix}$$

   (b) For the given matrix A find the eigenvalue that belongs to the eigenvector $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$ $\boxed{\text{1P}}$

$$A = \begin{pmatrix} -7 & 1 & 3 \\ 10 & 2 & -3 \\ -20 & -14 & 1 \end{pmatrix}$$